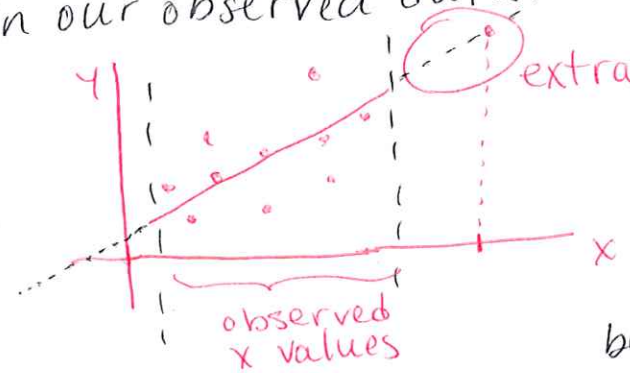
Last time: we continued studying simple linear regression, and how to use observed data to estimate the true, unknown population relationship between our predictor, X, and response, Y.

X: predictor
Y: response

population relationship unknown → estimate from data

We also discussed how to interpret the regression line that we've estimated. We should be cautious about using the fitted regression line to make conclusions about the relationship btwn X & Y far outside the range of X values in our observed data. This would be "extrapolation".



We discussed this in the context of the ~~vocabulary~~ language acquisition in the study. Infants in the study heard between 1,000 and 12,000 words/day. Using the fitted line to make predictions for $x = 0$ or $x = 50{,}000$ would be risky/unwise.

We also talked about the interpretation of the slope of the regression line. It is important not to suggest a causal relationship. For instance, we wouldn't say "increasing SAT score by 100 points tends to increase GPA by 0.25 points." Instead, it is better to think of the regression slope as a "difference."

β_1: NOT "change in y due to 1 unit change in X"

yes: "difference in y values for observations with 1 unit difference in X"

For example, "on average, we expect students whose SAT scores differed by 100 pts to have GPAs that differ by 0.25 pts."

So far, we have made a few key assumptions:

① We're assuming the true relationship really is linear

linearity $Y = \beta_0 + \beta_1 X + \varepsilon$

② We're also assuming that the random error terms are zero, on average.

$E(\varepsilon) = 0$

We collect data and find the regression line that estimates the true relationship between X and Y.

Under those assumptions, the least squares estimators of the slope and the intercept are "unbiased," meaning that their expected value equals the true value.

Consequence of ① + ②: $E(\hat{\beta}_1) = \beta_1$, $E(\hat{\beta}_0) = \beta_0$ unbiased

It makes sense to talk about their expected value, because the estimators are random variables.

$\hat{\beta}_1, \hat{\beta}_0$ random

↳ different values depending on randomly sampled data

Suppose we collected sample data, fit a regression line, and recorded the slope & intercept. If we went out and collected a NEW random sample of data, we could fit a new regression line, and the slope and intercept would be different.

We looked at a computational exercise previously, simulating

drawing 20 different sets of sample data & fitting 20 different regression lines. (3)

previous simulation exercise: 20 sample data sets, 20 values of $\hat{\beta}_0, \hat{\beta}_1$

In this way, it's the same as a sample proportion or a sample mean. There is some true population quantity, and the estimate we get from observed data will vary somewhat, depending on what values we happen to observe.

recall: $\hat{p}$ and $\bar{X}$ were also random

Because our estimators are unbiased, we expect to get the right values on average. We aren't estimating in a way that's systematically skewed away from the truth in one direction.

unbiased → $E(\hat{\beta}_0) = \beta_0$ → our estimates are correct, on average

In the simulation exercise fitting 20 different regression lines, if we took the average of all 20 slopes, it should be very close to the slope in the true population model.

Next, we will explore additional ways we can use linear regression models, if we are willing to make additional assumptions. (in addition to ① and ②, linearity and mean-zero error terms)

The extra assumptions we will make are about the error terms in the population model

pop. model $Y = \beta_0 + \beta_1 X + \varepsilon$

new assumptions: ε_i's normally distributed

ε_i's indep. of each other

ε_i's same variance

We will assume the error terms are normally distributed with variance σ^2, and that they are independent of each other.

$\varepsilon_i \overset{iid}{\sim} N(0, \sigma^2)$ (stated concisely)

What are the consequences of this?

* The y value for any particular value of X is also normally distributed.

Consequence: $Y|X=x \sim N(\beta_0 + \beta_1 x, \sigma^2)$ ($\beta_0 \to 1$, $\beta_1 \to 0.002$)

X = SAT score
Y = GPA

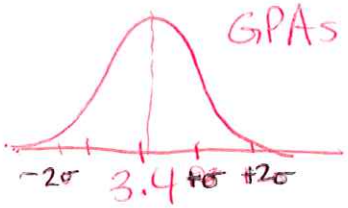
} GPAs of students w/1200 points on SAT

$Y|X=1200 = 3.4 + \varepsilon_i$

$\sim N(3.4, \sigma^2)$

In the example studying the relationship between SAT scores and 1st year GPA, if the point on the population regression line $\hat{}$ for X = 1200 is (1200, 3.4), then students with SAT scores of 1200 ~~have~~, on average have GPAs of 3.4.

But if we observed many students with that SAT ~~score~~ score, they would have different GPAs. And, their GPAs would be spread out above and below 3.4 according to a normal distribution.



We know how to generate random samples from a Normal distribution, so we can write down a population model and simulate the process of drawing sample data from the population.

~ examples in notebook ~

In general, when the variance of the error terms is larger, the data points tend to be more spread out around the line, and that makes it harder to estimate the true relationship

σ^2 large → y values more spread out @ any particular x value
↳ harder to estimate β_0, β_1

The same thing was true when we studied sample means. If X had variance σ^2, then the variance of the sample mean $\bar{X}$ was σ^2/n

recall: sample means
$\text{var}(X) = \sigma^2 \rightarrow \text{var}(\bar{X}) = \sigma^2/n$

When the population distribution was more spread out, so was the distribution of the estimator, $\bar{X}$.

Back to regression...
We have an estimator of the slope. We know it's unbiased. What is its variance? We would like to know the variance, to know how precise our estimator tends to be.

$\mathbb{E}(\hat{\beta}_1) = \beta_1$ $\quad \text{var}(\hat{\beta}_1) = ?$ How precise are estimates?

$$\text{var}(\hat{\beta}_1) = \frac{\sigma^2}{\sum_{i=1}^{n}(x_i - \bar{x})^2}$$

This expression agrees with what we were just talking about. If σ^2, the variance of the error terms, is larger, then the variance of the slope estimator is larger.

$\text{var}(\hat{\beta}_1)$ larger when σ^2 is larger

It's harder to precisely estimate the true slope when the data points are more spread out around the line.

what else can we learn from the expression for ~~var~~ $\text{var}(\hat{\beta}_1)$?

rearrange: $\sum_{i=1}^{n}(x_i-\bar{x})^2 = (n-1)\frac{1}{n-1}\sum_{i=1}^{n}(x_i-\bar{x})^2$

$= (n-1)s_x^2$ ← sample variance

$$\text{var}(\hat{\beta}_1) = \frac{\sigma^2}{(n-1)s_x^2}$$

Having $n-1$ in the denominator means that when we observe a lot of data points, so n is large, the variance of the slope estimator is small

$n-1$ in denominator → $\text{var}(\hat{\beta}_1)$ small when n large

↳ more data allows greater precision

when we observe more data, our estimates of the slope tend to be closer to the true value.

This is consistent with what we saw when studying sample means & sample proportions. As we took larger and larger samples, it became harder for the sample mean (or proportion) to be far away from the true population value

recall: same thing was true for $\bar{x}$ and $\hat{p}$

We can also see from the expression for the variance of $\hat{\beta}_1$ that the variance is smaller when the sample variance of the observed x values is large.

large sample variance of x values $\rightarrow$ smaller var($\hat{\beta}_1$)

Knowing the expected value and variance isn't enough for us to determine the probability of seeing some particular value of the slope estimator. In general, knowing expected value & variance doesn't tell us the probabilities of the possible outcomes.

When we started today, we chose to assume the error terms were normally distributed.

$\varepsilon_i \overset{iid}{\sim} N(0, \sigma^2)$

And another consequence of that assumption is that the estimators of slope and intercept are also normally distributed.

$\hat{\beta}_1 \sim N(\beta_1, var(\hat{\beta}_1))$

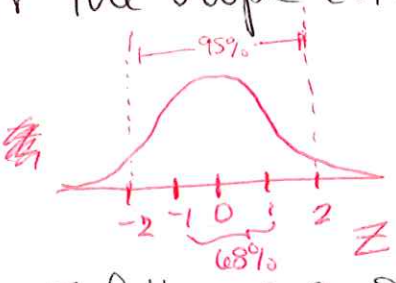
If we collected many different data sets, and fit a regression line for each of those data sets, we would get a bunch of slightly different values of the estimated slope. And those values would look like samples from a normal distribution.

$X \sim N(\mu, \sigma^2) \Rightarrow \frac{X-\mu}{\sigma} \sim N(0,1)$ (recall from homework)

In a homework problem, you confirmed that taking a normally distributed random variable, subtracting its mean, and dividing by its standard deviation, produces a standard normal variable (ie mean=0, variance=1).

We can do the same thing for the slope estimator:

$$Z = \frac{\hat{\beta}_1 - \beta_1}{\sqrt{\text{var}(\hat{\beta})}} \sim N(0,1)$$



We'll call this quantity Z. Z follows a standard normal dist. It takes values between -1 and +1 about 68% of the time, and between -2 and +2 about 95% of the time.

But the expression for Z includes σ^2, the variance of the error terms. That variance, σ^2, is a feature of the true population model, which is unknown.

$$Z = \frac{\hat{\beta}_1 - \beta_1}{\sqrt{\frac{\sigma^2}{(n-1)S_x^2}}}$$

don't know σ^2 → estimate it from data

The error term ε_i is the vertical distance from data point (x_i, y_i) to the true population regression line.

$$\varepsilon_i = y_i - (\beta_0 + \beta_1 x_i)$$

(β_0, β_1: unknown)

$$\hat{e}_i = y_i - (\hat{\beta}_0 + \hat{\beta}_1 x_i)$$

($\hat{\beta}_0 + \hat{\beta}_1 x_i = \hat{y}_i$)

We don't know the value of that error term, but we do know the value of the residual, $\hat{e}_i$. The residual is the vertical distance from the data point (x_i, y_i) to the estimated regression line.

The residuals are estimates of the true error terms,

$\hat{e}_i$ is an estimate of true ε_i

so we can use the sample variance of the residuals to approximate σ^2, the variance of the error terms.

$$\sigma^2 \approx \frac{1}{n-2} \sum_{i=1}^{n} (\hat{e}_i - \bar{\hat{e}})^2$$ sample variance

For least squares regression, the sum of the residuals always equals zero*, so the average of the residuals = 0.

$\sum_{i=1}^{n} \hat{e}_i = 0$ always, so $\bar{\hat{e}} = 0$ also

$$\sigma^2 \approx \frac{1}{n-2} \sum_{i=1}^{n} \hat{e}_i^2 = \hat{\sigma}^2$$

$\sqrt{\hat{\sigma}^2}$: also called "residual standard error" or "standard error of residuals"

* we did not prove this

Notice that when calculating standard error of the residuals, we divided by $n-2$ instead of $n-1$. We will discuss the reason for that later on, when we study regression models with multiple predictor variables.

$\hat{\sigma}^2 = \frac{1}{n-2}\sum_{i=1}^{n} \hat{e}_i^2$ but $S_x^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2$

So, instead of being able to directly calculate Z, we approximate it by plugging in the residual standard error $\hat{\sigma}^2$.

$$Z = \frac{\hat{\beta}_1 - \beta_1}{\sqrt{\frac{\sigma^2}{(n-1)S_x^2}}} \xrightarrow{\text{substitute } \hat{\sigma}^2} t = \frac{\hat{\beta}_1 - \beta_1}{\sqrt{\frac{\hat{\sigma}^2}{(n-1)S_x^2}}}$$

Similarly to what we studied for confidence intervals and hypothesis tests about an unknown population mean μ, substituting an estimate of σ^2 means that we need to refer to the t distribution instead of the normal distribution.

$Z \sim N(0,1)$ but $t = \frac{\hat{\beta}_1 - \beta_1}{\sqrt{\frac{\hat{\sigma}^2}{(n-1)S_x^2}}} \sim t_{n-2}$ (because of using estimated $\hat{\sigma}^2$)

$df = n-2$

Specifically, this quantity follows a t distribution with $n-2$ degrees of freedom.